

## ReactJS HOL 14 – Using React Context API

In this hands-on exercise, I implemented the React Context API to manage the application theme. A ThemeContext was created and provided to all components using ThemeContext.Provider. The selected theme was passed through Context instead of props, and the EmployeeCard component accessed the theme using the useContext hook. The application successfully switched between Light and Dark themes.

### Source Code

#### 1. ThemeContext.js

```
import React from "react"

const ThemeContext = React.createContext();

export default ThemeContext;
```

#### 2. App.js

```
import './App.css';

import { EmployeesData } from './Employee';
import EmployeesList from './EmployeesList';

import { useState } from 'react';

import ThemeContext from './ThemeContext';

function App() {

  const Employees = EmployeesData;

  const [theme, setTheme] = useState('light');

  return (

    <ThemeContext.Provider value={theme}>

      <div>

        <label>SELECT A THEME </label>

        <select

          value={theme}

          onChange={(e) => setTheme(e.target.value)}

        >

          <option value="light">Light</option>
```

```

        <option value="dark">Dark</option>
      </select>

      <EmployeesList employees={Employees} />
    </div>
  </ThemeContext.Provider>
);
}

export default App;

```

### 3. EmployeesList.js

```

import EmployeeCard from './EmployeeCard';

function EmployeesList(props) {
  return (
    <div>
      <h1>Employees List</h1>
      {
        props.employees.map(employee => (
          <EmployeeCard
            employee={employee}
            key={employee.id}
          />
        ))
      }
    </div>
  );
}

export default EmployeesList;

```

### EmployeeCard.js

```

import { useContext } from 'react';
import ThemeContext from './ThemeContext';

```

```

import Styles from './EmployeeCard.module.css';

function EmployeeCard(props) {

  const theme = useContext(ThemeContext);

  return (

    <div className={Styles.Card}>

      <h3>{props.employee.name}</h3>

      <p>{props.employee.email}</p>

      <p>{props.employee.phone}</p>

      <p>

        <a href="#" className={theme}>Edit</a>

        <a href="#" className={theme}>Delete</a>

      </p>

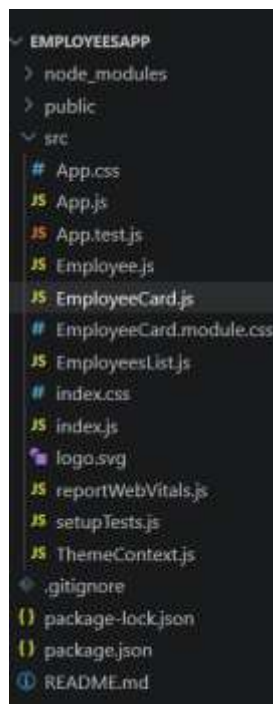
    </div>

  );
}

export default EmployeeCard;

```

**Figure 1:** Project Structure



**Figure 2: Browser output-Light Theme**



**Figure 3: Browser output=Dark Theme**



**Figure 4: Terminal output**

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS E:\Users\JSP\OneDrive\Documents\employeesapp> npm start
assets by path 2.22 KiB
  asset index.html 1.67 KiB [emitted]
  asset asset-manifest.json 458 bytes [emitted]
cached modules 1.37 KiB (javascript) 31.3 KiB (runtime) [cached] 126 modules
./src/EmployeeCard.js 2.79 KiB [built]

WARNING in src/EmployeeCard.js
  Line 15:17: The href attribute requires a valid value to be accessible. Provide a valid, navigable address as the href value. If you cannot provide a valid href, but still need the element to resemble a link, use a button and change it with appropriate styles. Learn more: https://github.com/jsx-eslint/eslint-plugin-jsx-ally/blob/HEAD/docs/rules/anchor-is-valid.md  jsx-ally/anchor-is-valid
  Line 16:17: The href attribute requires a valid value to be accessible. Provide a valid, navigable address as the href value. If you cannot provide a valid href, but still need the element to resemble a link, use a button and change it with appropriate styles. Learn more: https://github.com/jsx-eslint/eslint-plugin-jsx-ally/blob/HEAD/docs/rules/anchor-is-valid.md  jsx-ally/anchor-is-valid

webpack 5.68.0 compiled with 1 warning in 73 ms
```

## Result

The React Context API was implemented successfully. The selected theme was shared across components using ThemeContext, and the EmployeeCard component accessed the theme using the useContext hook. The application successfully switched between Light and Dark themes without using prop drilling.